

Tutorial 1: Kinematics and Relative Velocity

1D Kinematics

Q1.

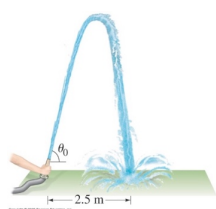
A rock is thrown vertically upward with a speed of 12.0 m/s . Exactly 1.00 s later, a ball is thrown up vertically along the same path with a speed of 18.0 m/s . (a) At what time will they strike each other? (b) At what height will the collision occur? (c) Answer (a) and (b) assuming that the order is reversed: the ball is thrown 1.00 s before the rock.

Kinematics and Calculus:

Q2.

Air resistance ($F_{\text{air}} = -bv$) acting on a falling body can be taken into account by the approximate relation for the acceleration, $a = dv/dt = g - kv$, where $k = b/m$ is a constant. (a) Derive a formula for the velocity of the body as a function of time assuming it starts from rest. [Hint: Change variables by $u = g - kv$] (b) Determine an expression for the terminal velocity, which is the maximum value of velocity.

2D Kinematics

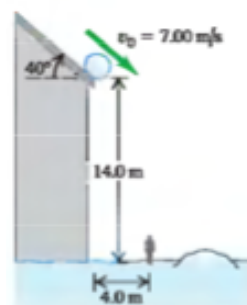


Q3.

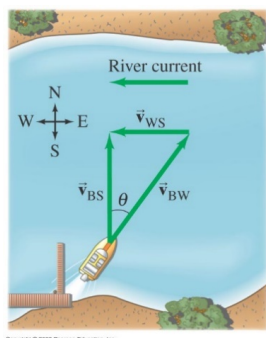
A fire hose held near the ground shoots water at a speed of 6.5 m/s . At what angle(s) should the nozzle point in order that the water land 2.5 m away? Why are there two different angles? Sketch the two trajectories.

Q4.

A ball rolls off a roof that slopes downward at an angle of 40° to the horizontal. The edge of the roof is 14.0 m above the ground, and the ball has a speed of 7.0 m/s as it rolls off the roof. Ignore air resistance. (a) How far from the edge of the building does the snowball strike the ground if it doesn't strike anything else while falling? (b) Draw x - t , y - t , v_x - t and v_y - t graphs for the motion in part (a). (c) A man 1.75 m tall is standing 4.0 m away from the edge of the building. Will the ball hit him?



Relative Velocity



Q5.

A motorboat whose speed in still water is 3.40 m/s must aim upstream at an angle of 19.5° (with respect to a line perpendicular to the shore) in order to travel directly across the stream. (a) What is the speed of the current? (b) What is the resultant speed of the boat with respect to the shore?

Q6. Challenging

A person stands at the base of a hill that is a straight incline making an angle ϕ with the horizontal. For a given initial speed v_0 , at what angle θ to the horizontal, should objects be thrown so that the distance d they land up the hill is as large as possible?



Answers

Q1. 1.45 s , 7.10 m , collision below ground, i.e. don't collide

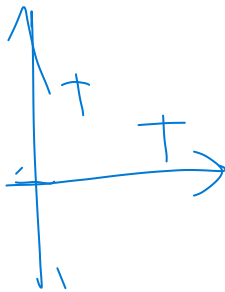
Q2. $v = (g/k)(1 - \exp(-kt))$, $v_{\text{max}} = g/k$

Q3. 1.13 m/s^2 , 3.20 m/s^2

Q4. (a) 1.29 s (c) No. Q5. 18° , 72° Q6. $\theta = \pi/4 + \phi/2$

Q1 a) Consider rock.
 $U_R = 12.0 \text{ m/s}$

$U_B = 18.0 \text{ m/s}$



t is defined as time after the rock is thrown.

Consider ~~vertical~~ S_y of both.
 using $S_y = U_y t + \frac{1}{2} a t^2$

$$-\frac{g}{2} t^2 + \frac{g}{2} (2t) - \frac{g}{2}$$

$$U_R t + \frac{1}{2} (-g) (t^2) = U_B t + \frac{1}{2} (-g) (t^2 - 2t + 1)$$

$U_R t - U_B t$

$$-\frac{g}{2} t^2 + \frac{g}{2} t^2 + (U_R - U_B - g) t + (U_B + \frac{g}{2}) = 0$$

$t = 1.45 \text{ s}$

b) At $t = 1.45$

$$S_R = U_R t + \frac{1}{2} a t^2$$

Sub $U_R = 12.0, a = -9.8, t = 1.45$
 $S_R = 7.10 \text{ m}$

c) ~~Use the same~~ Consider the following.

$$U_R t + \frac{1}{2} (-g) (t^2) = U_B t + U_B + \frac{1}{2} (-g) (t^2 - 2t + 1)$$

$$-\frac{g}{2} t^2 + \frac{g}{2} t^2 + (U_R - U_B + g) t + (-U_B + \frac{g}{2}) = 0$$

$t = 3.45 \text{ s}$

At $t = 3.45$

$$S_R = U_R t + \frac{1}{2} (-g) (t^2) = -16.9 \text{ m}$$

Height -ve, below ground will not occur.

Q2.

Fair $= -bv$

$a = \frac{dv}{dt} = g - kv$, $k = \frac{b}{m}$ is a constant.

a) Consider start from rest

$a = g - kv$

$v = 0 \Rightarrow \downarrow g \downarrow a = g$

$v = 0 \Rightarrow \downarrow g \downarrow a = g - kv$

$v = v_{\text{max}} \Rightarrow \downarrow g \downarrow a = g - kv$

$$\int_0^v \frac{1}{g - kv} dv = \int_0^t dt$$

Sub $u = g - kv$

$$\frac{du}{dv} = -k \Rightarrow du = -k dv$$

when $v = 0, u = g$

$v = v, u = g - kv$

$$\int_g^{g-kv} \frac{1}{u} du = \int_0^t dt$$

$$\frac{1}{-k} \left[\ln|u| \right]_g^{g-kv} = t$$

$$\frac{1}{-k} (\ln|g - kv| - \ln|g|) = t$$

$$\frac{1}{-k} \ln \left| \frac{g - kv}{g} \right| = t$$

$$\ln \left| \frac{g - kv}{g} \right| = -kt$$

$$\frac{g - kv}{g} = e^{-kt}$$

$$g - kv = g e^{-kt}$$

$$v = \frac{g}{k} (1 - e^{-kt})$$

b) ~~As~~ max velocity occurs when $\frac{dv}{dt} = 0$ at $t = \infty$
 $v = \frac{g}{k}$

3.

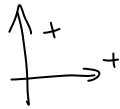
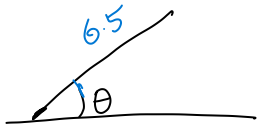
~~Speed = 6.5 m/s~~

$$U = 6.5$$

$$U_x = 6.5 \cos \theta$$

$$U_y = 6.5 \sin \theta$$

Land 2.5m away $\Rightarrow S_x = 2.5m$ when $S_y = 0m$



~~6.5 \cos \theta t + \frac{1}{2} (-9.8) t^2 = 2.5~~ --- (1)

$$6.5 \sin \theta t + \frac{1}{2} (-9.8) t^2 = 0 \text{ --- (2)}$$

From (2) $t * (6.5 \sin \theta - \frac{9.8}{2} t) = 0$

~~t =~~

~~t = 0~~ req since initial

$$6.5 \sin \theta - \frac{9.8}{2} t = 0$$

$$t = \frac{2(6.5 \sin \theta)}{9.8} \text{ --- (2)}$$

Sub (2) into (1)

$$\frac{(2)}{2.5} \frac{(6.5)^2}{9.8} \sin \theta \cos \theta = 1$$

$$0 \leq \theta \leq 90^\circ$$

$$0 \leq 2\theta \leq 180^\circ$$

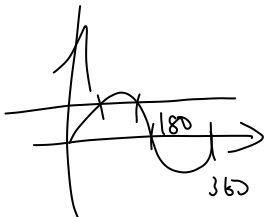
Given $\sin(2\theta) = 2 \sin \theta \cos \theta$

~~(6.5)~~

$$\sin(2\theta) = \frac{1(2.5)(9.8)}{(6.5)^2}$$

$$\frac{0.309}{2\pi} \times 360^\circ$$

$$\frac{1.26}{2\pi} \times 360^\circ$$

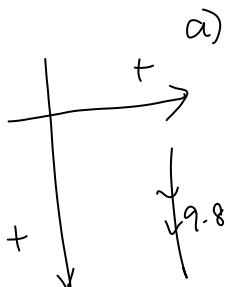


$$2\theta = \sin^{-1} \left(\frac{1(2.5)(9.8)}{(6.5)^2} \right)$$

$$2\theta = 35.44^\circ, 180^\circ - 35.44^\circ$$

$$= 18^\circ, 72^\circ$$

Q4.



a)

$$7 \cos 40^\circ$$

$$7 \sin 40^\circ$$

Consider ~~x-component~~ y -component $S_y = U_y t + \frac{1}{2} a_y t^2$

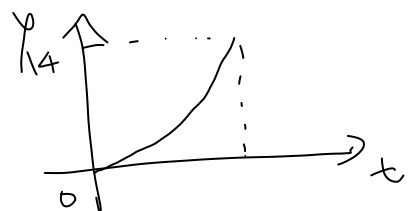
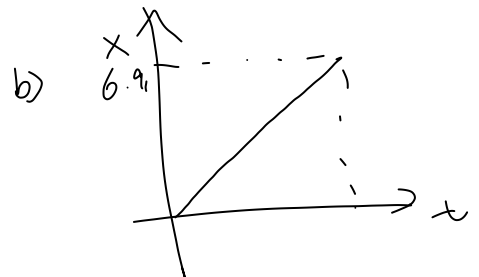
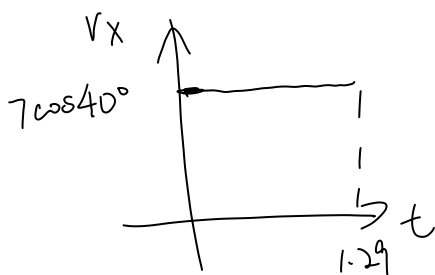
$$S_y = 14.0 = 7 \sin 40^\circ t + \frac{1}{2} (-9.8) t^2$$

Solve for t : $t = 1.29$ or -2.1 (neg; before do solving)

Consider x -component,

$$S_x = U_x t = (7 \cos 40^\circ) (1.29)$$

$$= 6.92$$



c) Consider S_y when $S_x = 4$

$$\text{When } S_x = 4.0, \quad t_1 = \frac{S_x}{u_x} = \frac{4.0}{7 \cos 40^\circ} = 0.746 \text{ s}$$

Consider S_y

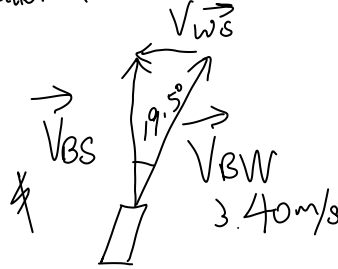
$$S_y = u_y t + \frac{1}{2} a t^2$$

$$= (7 \sin 40^\circ)(0.746) + \frac{1}{2}(-9.8)(0.746)^2 = -6.0836 \text{ m}$$

$$\text{Height from ground} = 14 - 6.0836 = 7.92$$

$\therefore$ taller than man $\rightarrow$ will not hit.

5.



$$\begin{aligned} \vec{V}_{BS} &= \vec{V}_{BW} \sin 19.5^\circ \\ &= 3.40 \cos 19.5^\circ \\ &= 3.20 \text{ m/s} \end{aligned}$$

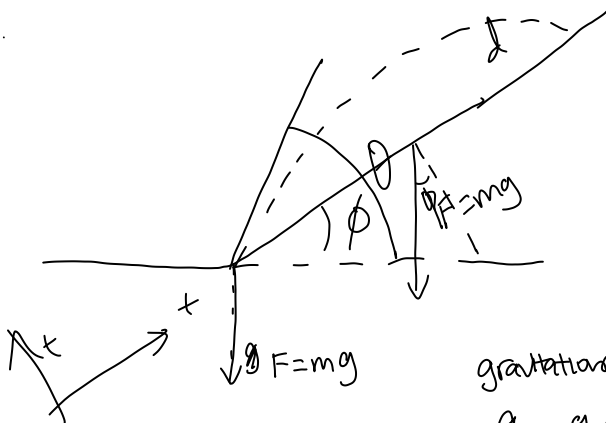
$$\vec{V}_{BS} = \vec{V}_{BW} + \vec{V}_{WS}$$

$$\vec{V}_{WS} = \vec{V}_{BS} - \vec{V}_{BW}$$

$$\begin{aligned} \vec{V}_{WS} &= -\vec{V}_{BW} \sin 19.5^\circ \\ &= -(3.10) \sin 19.5^\circ \\ &= -1.13 \text{ m/s} \end{aligned}$$

The speed is 1.13 m/s

6.



~~Grav~~

y-component

$\hookrightarrow$ perpendicular to slope

x-component

$\hookrightarrow$ parallel to slope

gravitational acceleration g

$$g_y = g \cos(\phi)$$

$$g_x = g \sin(\phi)$$

Consider motion along the slope

$$V_y = V_0 \sin(\theta - \phi) \quad V_x = V_0 \cos(\theta - \phi)$$

Land on the hill $\rightarrow S_y = 0$

$$S_y = V_y t + \frac{1}{2} a t^2$$

$$V_0 \sin(\theta - \phi) t + \frac{1}{2} (-g \cos \phi) t^2 = 0$$

$$t(V_0 \sin(\theta - \phi) - \frac{1}{2} g \cos \phi t) = 0$$

$\hookrightarrow$ scenario ~~where~~ initial

$$t = \frac{2 V_0 \sin(\theta - \phi)}{g \cos \phi} \quad \text{--- (1)}$$

Go to x-displacement

$$S_x = V_x t + \frac{1}{2} a_x t^2 = V_0 \cos(\theta - \phi) t + \frac{1}{2} (-g \sin \phi) t^2 \quad - (2)$$

Sub ① into ②

$$S_x = \frac{2V_0^2 \cos(\theta - \phi) \sin(\theta - \phi)}{g \cos \phi} + \frac{-\frac{1}{2} g \sin \phi \left(\frac{2V_0^2 \sin^2(\theta - \phi)}{g^2 \cos^2 \phi} \right)}{g \cos \phi}$$

$$S_x = \frac{2V_0^2}{g \cos \phi} \left[\sin(\theta - \phi) \cos(\theta - \phi) - \frac{g \sin \phi}{g \cos \phi} \sin^2(\theta - \phi) \right]$$

$$S_x = \left[\frac{2V_0^2 \sec \phi}{g} \right] \left[\frac{\sin(2\theta - 2\phi)}{2} - \tan \phi \sin^2(\theta - \phi) \right]$$

Recall

$$\frac{d}{dx} [f(x)]^n = n[f(x)]^{n-1} f'(x)$$

$$\frac{dS_x}{d\theta} = \left[\frac{2V_0^2 \sec \phi}{g} \right] \left[\frac{1}{2} \cos(2\theta - 2\phi) - 2 \tan \phi \sin(\theta - \phi) \cos(\theta - \phi) \right]$$

$$\frac{dS_x}{d\theta} = \left[\frac{2V_0^2 \sec \phi}{g} \right] \left[\cos(2\theta - 2\phi) - 2 \tan \phi \sin(\theta - \phi) \cos(\theta - \phi) \right]$$

$$\frac{dS_x}{d\theta} = \left[\frac{2V_0^2 \sec \phi}{g} \right] \left[\cos(2\theta - 2\phi) - 2 \tan \phi \sin(\theta - \phi) \cos(\theta - \phi) \right]$$

furthest distance $\Rightarrow \frac{dS_x}{d\theta} = 0$

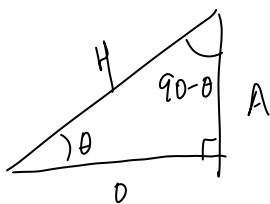
$$\frac{2V_0^2 \sec \phi}{g} [\cos(2\theta - 2\phi) - \tan \phi \sin(2\theta - 2\phi)] = 0$$

Since $\sec \phi \neq 0$ for any θ value

$$\cos(2\theta - 2\phi) - \tan \phi \sin(2\theta - 2\phi) = 0$$

$$\cos(2\theta - 2\phi) = \tan \phi \sin(2\theta - 2\phi)$$

$$\cot(2\theta - 2\phi) = \tan \phi$$



$$\tan \theta = \frac{D}{A}$$

$$\cot \theta = \frac{1}{\tan \theta} = \frac{A}{D}$$

$$\cot(90 - \theta) = \frac{A}{D} = \tan \theta$$

$$\tan(90 - \theta) = \frac{A}{D}$$

~~tan~~

~~tan~~

$$\tan\left(\frac{\pi}{2} - 2\theta + 2\phi\right) = \tan \phi$$

$$\frac{\pi}{2} - 2\theta + 2\phi = \phi$$

$$\frac{\pi}{2} + 2\phi - \phi = 2\theta$$

$$\frac{\pi}{2} - 2\theta = \frac{\pi}{2} + \phi$$

$$\theta = \frac{\pi}{4} + \frac{\phi}{2}$$

Presentation

- ① Clearly drawn coord system (direction)
- ② Keep analytical form as far as ~~to~~ possible.